

Group 3 Draft Design Document

Executive Summary

Table of Contents

- Executive Summary
- Table of Contents
- Introduction
- Design
- Evaluation
- Appendix 1 – Problem Formulation
- Appendix 2 – Planning
- Appendix 3 – Test Plan & Results
- Appendix 4 – Review
- GPS and Telemetry
- Communication Protocol
- Mobile App Architecture
- Developer Onboarding
- Dual BLE + Wi-Fi Operation
- Camera Integration
- Server Infrastructure

Introduction

Need & Goal Statements

Need Statement

There is a need to simplify the process of piloting a drone by removing the requirement for continuous manual control.

Goal Statement

The goal of this project is to design an autonomous drone that follows a user without requiring continuous manual control.

Design Objective

The objective of this project is to create a pilotless autonomous drone that reduces manual control of piloting by 90%.

Personas

Persona 1: YouTuber

Name: Alex

Age: 25

Occupation: YouTube Content Creator

Alex produces travel and lifestyle videos for YouTube and often films alone while visiting new locations. Alex wants to include moving shots and aerial perspectives to make videos more engaging, but using a tripod limits the ability to capture dynamic footage. A smartphone-controlled drone

would allow Alex to move the camera while appearing in the video, making it easier to create professional-looking content without needing another person to operate the camera.

Persona 2: Musician

Name: Maya

Age: 28

Occupation: Independent Musician

Maya records live performances and music videos for social media platforms. Most recordings are done independently with limited equipment, which restricts camera movement and creative angles. With a drone controlled through a smartphone, Maya could capture overhead shots or move the camera during performances to create more visually engaging music videos.

Persona 3: Movie Producer

Name: Daniel

Age: 34

Occupation: Independent Film Producer

Daniel works on small independent film projects with a limited production crew. Capturing aerial or moving shots often requires additional equipment or specialized drone operators, which increases production costs. A smartphone-controlled drone would allow Daniel to capture establishing shots and dynamic camera movement more easily, providing greater flexibility during filming without requiring extra personnel.

Research into existing designs

Several existing drone and camera systems provide aerial recording capabilities, but many are not optimized for simple smartphone-based operation by a single individual.

DJI Mini Series The DJI Mini 3 and Mini 4 drones are popular consumer drones that offer high-quality cameras and stable flight systems. These drones are capable of recording high-resolution video and include automated flight modes such as tracking and waypoint navigation. However, DJI drones typically require a dedicated remote controller for reliable operation. While some smartphone control options exist, the systems are designed primarily for users with some experience flying drones. The learning curve and controller hardware may discourage casual users who only need a simple filming solution.

HoverAir X1 The HoverAir X1 is a compact drone designed specifically for quick personal recording. It can automatically follow a subject and capture video without requiring manual control. This product is aimed at casual users who want simple aerial footage. While this system simplifies filming, it relies heavily on automated tracking modes rather than manual directional control. Users have limited ability to control precise drone movement or adjust camera position in real time.

Ryze Tello The Ryze Tello drone is a small drone that can be controlled through a smartphone app using Wi-Fi communication. The app includes on-screen controls that allow users to move the drone in different directions. Although Tello demonstrates the feasibility of smartphone-based drone control, it is primarily designed as an educational or beginner drone. The camera quality and control responsiveness are limited compared to higher-end recording systems.

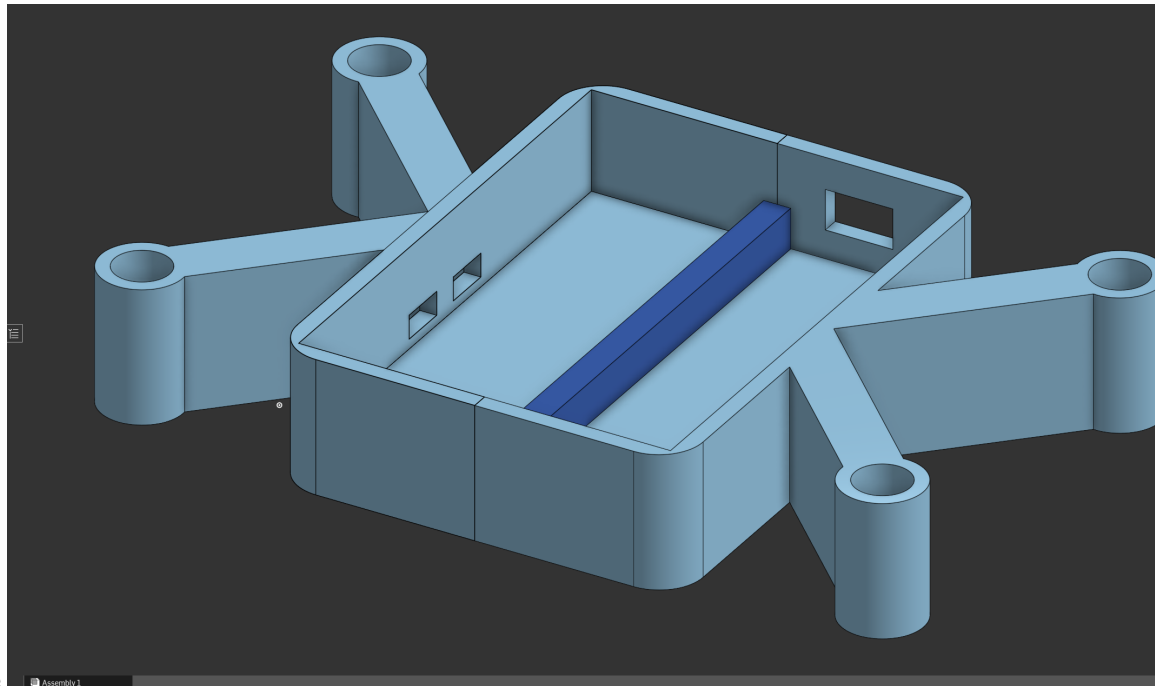
Sustainability Statement

The proposed drone system considers sustainability through efficient energy use, reusable components, and minimizing unnecessary hardware.

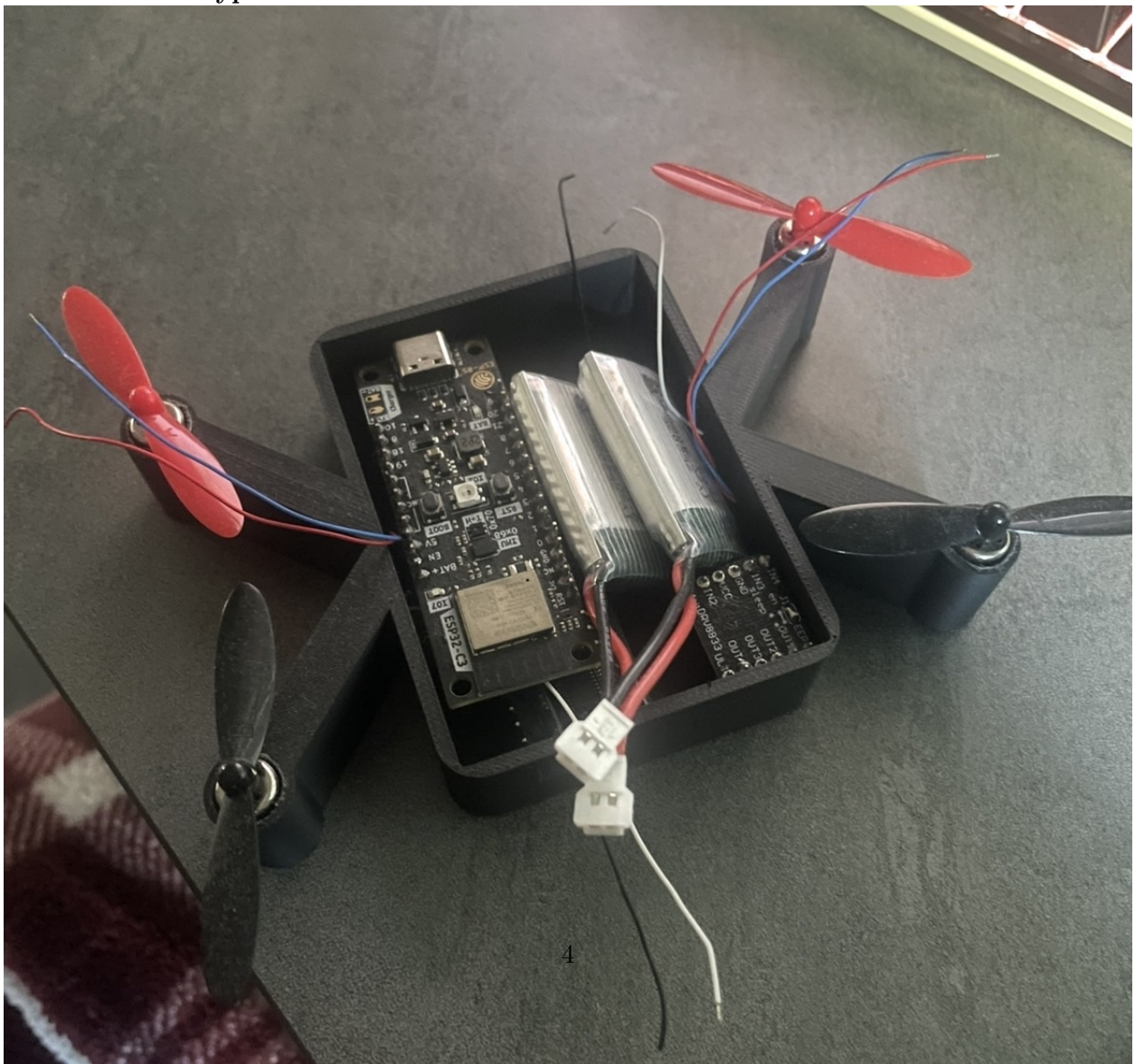
The drone will operate using rechargeable batteries, allowing repeated use without generating disposable waste. Efficient electronic components and lightweight materials will help reduce power consumption and extend flight time. The design will also emphasize repairability and replaceable components, such as propellers, batteries, and camera modules. This approach extends the lifespan of the device and reduces electronic waste. Additionally, the system uses the user's existing smartphone as the control interface, eliminating the need for a separate controller and reducing additional hardware production.

Through these design considerations, the system aims to provide a functional product while reducing environmental impact and encouraging long-term usability.

Design



Aesthetic Prototype

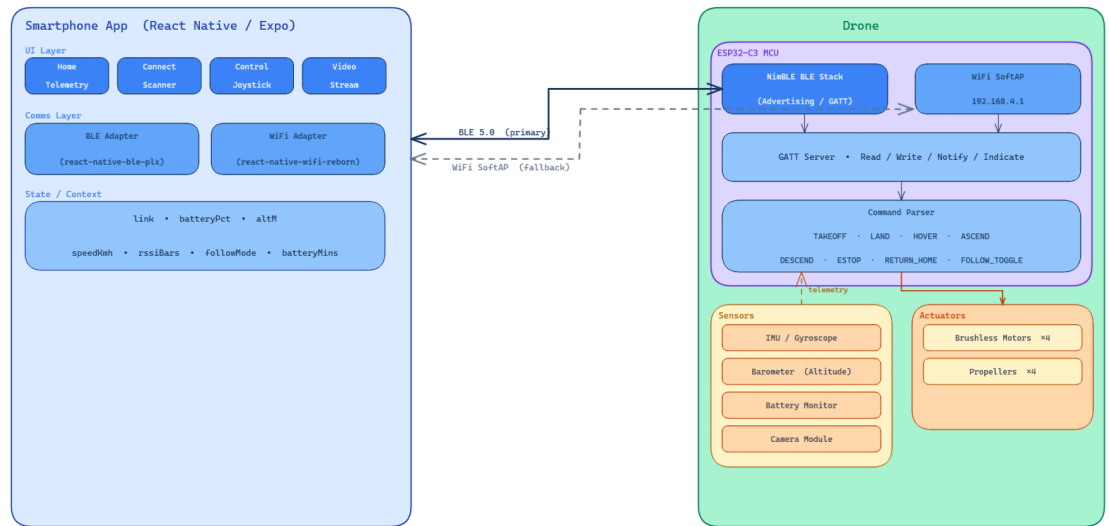


Design for Manufacture, Assembly, Maintenance

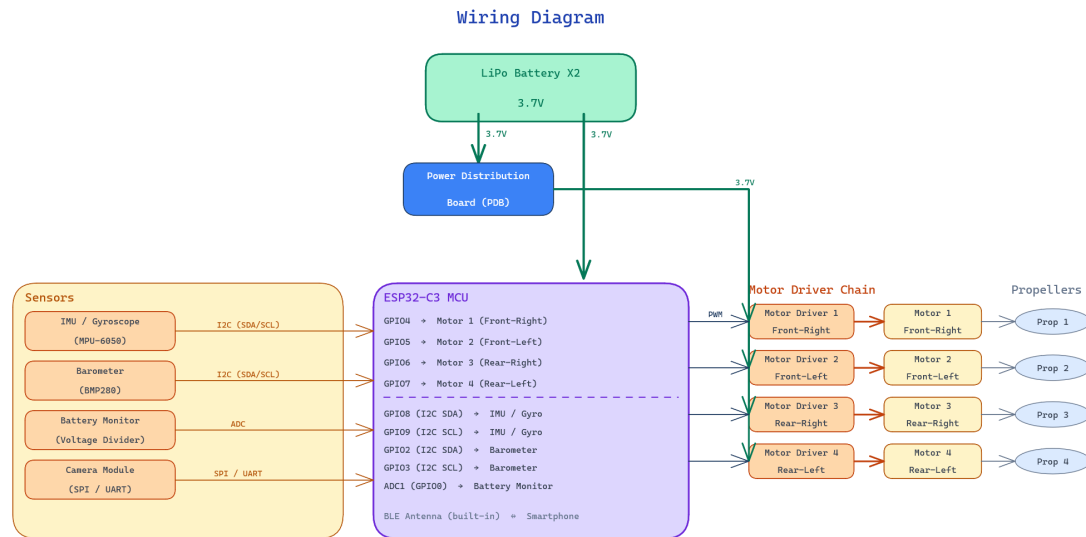
The drone frame is designed with manufacturability and assembly in mind. The structure uses a lightweight 3D-printed frame that can be produced quickly using common additive manufacturing equipment. The design separates the frame into simple components so that damaged parts, such as propeller arms or mounts, can be replaced individually without rebuilding the entire structure. Standard fasteners and mounting holes are used to attach motors, electronics, and the battery to ensure compatibility with commonly available hardware.

Assembly focuses on a modular approach where each subsystem—power, flight control, propulsion, and camera—can be installed independently. This modularity simplifies both initial construction and troubleshooting during development. Wiring is routed through the frame to reduce interference with the propellers and to keep the system organized and accessible.

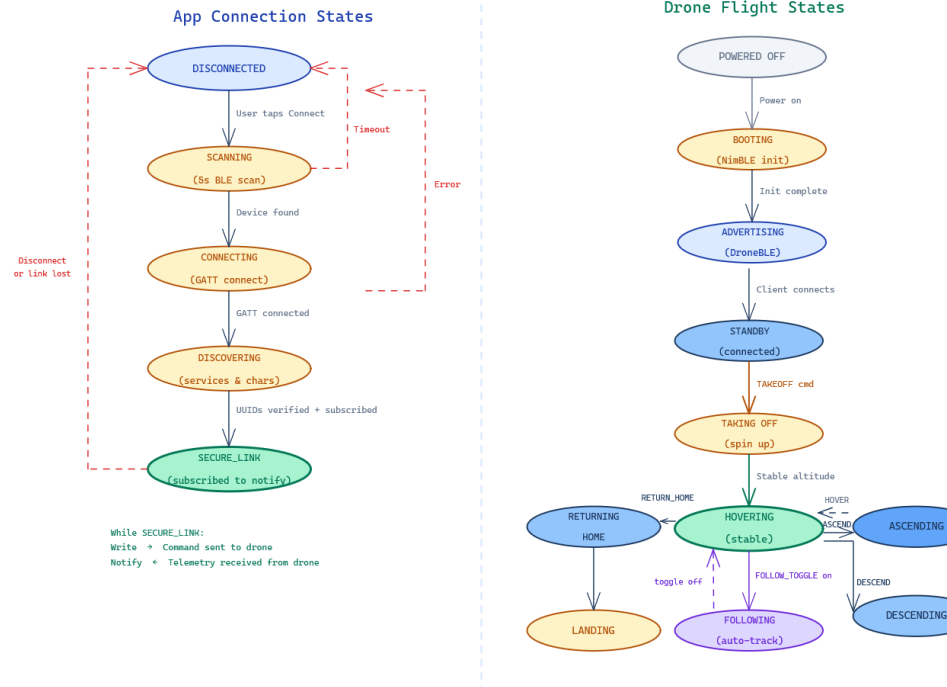
Maintenance was also considered during the design process. Key components such as the battery, propellers, and flight-controller PCB are easily accessible so they can be replaced or serviced when needed. Because drones experience mechanical stress during flight, the design allows quick replacement of high-wear parts, helping extend the lifespan of the system and reduce downtime during operation.



Block Diagrams



Wiring Diagrams



State Transition Diagrams

Technology

The finished product centers on a **custom flight-controller PCB** as the computing platform. The board integrates wireless communication (Bluetooth Low Energy and Wi-Fi), sufficient processing for sensor fusion and flight control, and GPIO interfaces for motor drivers and peripherals. Integrated BLE enables low-latency command and status exchange with the mobile app; integrated Wi-Fi provides a local soft access point for high-bandwidth telemetry and video without requiring internet connectivity.

The propulsion system consists of four brushless motors with propellers in a quadcopter configuration. Each motor is driven independently through electronic speed control, allowing differential thrust for lift, attitude, and translation. A rechargeable lithium-polymer battery supplies power with separate rails for logic and motor current as required by the PCB design.

Onboard sensing (product)

Sensor	Role
IMU	Attitude estimation and closed-loop flight control
GNSS	Outdoor position for navigation and follow-me
Magnetometer	Heading for yaw alignment during follow
Barometer	Altitude hold and vertical state (product firmware)

A fixed camera module is part of the product concept for autonomous filming; the mobile app includes a video tab prepared for an MJPEG or similar stream served over the drone’s local Wi-Fi link.

User interface

A **React Native mobile application** (four tabs: Home, Connect, Control, Video) is the sole operator interface. The user’s smartphone provides GNSS for follow-me geometry, displays combined phone and drone position, and sends commands over a **dual wireless link**: secure Wi-Fi for telemetry (and future video) and BLE for low-latency flight commands. This removes the need for a dedicated radio controller.

Autonomous follow-me (product behavior)

The product goal is hands-off filming: the user walks; the drone maintains a safe offset using **phone GNSS** (subject) and **drone GNSS + compass** (aircraft state). The mobile app computes navigation intent (distance, bearing, arrival) and maps it to a documented binary command set (NAV_* opcodes). The flight controller enforces speed limits, geofencing, and failsafes on board—control logic is not safety-critical on the phone alone.

Flight control architecture

The diagram shows the **two-layer control model**: the phone sends follow intent over BLE (NAV_* opcodes); an onboard navigation executor (product design) translates intent into attitude or velocity setpoints for a **100 Hz inner loop** (IMU → complementary-filter attitude estimate → PID → X-quad motor mix). Solid lines indicate paths demonstrated on the bench; dashed lines indicate designed behavior not yet integrated on the prototype flight-controller image.

Simulations

Before and alongside flight testing, control and sensor algorithms are validated in **hardware-in-the-loop (HIL)** simulation: the same PID and motor-mixing logic runs against a software plant so tuning and regressions do not always require a tethered airframe. Physical prototypes remain essential for comms, wiring, and integration, but HIL reduces risk when extending autonomous modes.

Note on current prototypes: Bench and demo units may use interim microcontroller modules and split firmware images while the custom PCB and unified production image

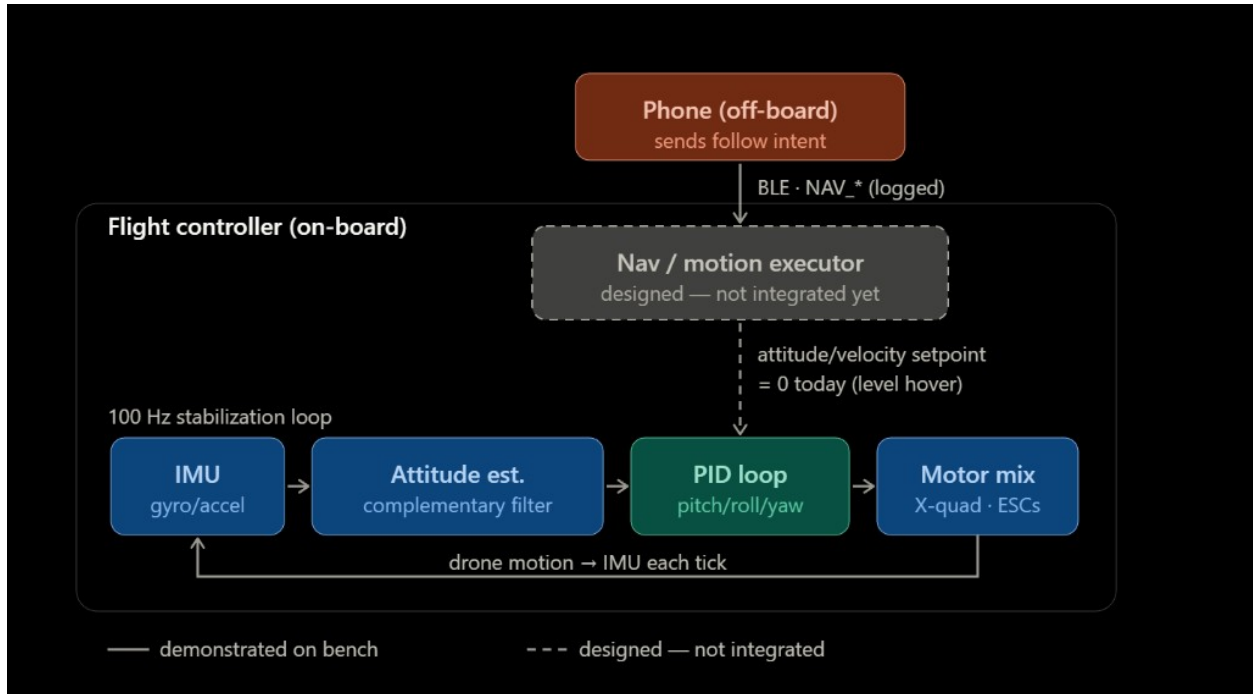


Figure 1: Two-layer flight control loop

are finalized. Behavior described in Sections 10–12 reflects the **intended product protocol and app architecture**, which the prototype app and firmware already implement at the protocol level.

Evaluation

This section separates **what the finished product is designed to do** from **what the current bench prototype has demonstrated**. The design targets a custom PCB, dual wireless links, and autonomous follow-me; prototype work validates communication, sensing, and partial motor control ahead of full flight integration.

Design validation (product intent)

The architecture is judged viable because:

- A mobile app can maintain a **secure local link** to the drone (Wi-Fi telemetry plus BLE commands) without cloud or internet dependency.
- A **shared command and telemetry protocol** (binary commands, JSON/TEL telemetry) is implemented end-to-end between app and firmware.
- **Phone and drone GNSS** can be shown together and fed into follow-to-phone navigation logic on the handset.
- **Manufacturing tests** (Appendix 3) define pass/fail criteria for assembly, power-on, comms, motor mapping, and tethered flight before shipment.

Prototype demonstrations (current quarter)

The bench prototype has shown:

Capability	Status
BLE discovery, connect, reconnect	Demonstrated
Wi-Fi join, TLS telemetry (<code>/gps</code> , WSS)	Demonstrated
Hybrid app: WSS link state + BLE-first commands	Demonstrated
Drone GNSS + compass in telemetry	Demonstrated (outdoor fix dependent)
BMP280 relative altitude in telemetry (<code>altM</code>)	Demonstrated (<code>wifi_gps_softap</code> bench)
PID / attitude loop on bench (<code>flight_control</code>)	Demonstrated (props off; IMU + PID logging)
Hardware-in-the-loop sim (<code>flight_sim</code> + Python bridge)	Demonstrated (desktop physics; optional)
Phone-side follow navigation + intent → <code>NAV_*</code> commands	Demonstrated (app → firmware path)
Closed-loop autonomous follow flight	Not yet — flight controller does not fully execute <code>NAV_*</code> on prototype firmware
Per-motor manual throttle over BLE in unified demo build	Limited — high-level bench demo uses scripted takeoff opcode
Live camera stream	Not yet — video endpoint is a placeholder stream
Custom PCB in loop	In progress — some demos use interim dev hardware

These results support the design without claiming every product feature is flight-ready.

Functional prototype (physical)

The physical prototype consists of a **3D-printed quadcopter frame**, four brushless motors with propellers, a LiPo power system, and a **flight-controller assembly** (custom PCB where available, otherwise interim modules wired to the same logical interfaces). Motors are driven through ESC hardware appropriate to the PCB revision (product design: independent channels per arm).

App screenshots

Screens reflect the four-tab app: connection management, telemetry and follow UI on Home, manual Control, and Video (stream when firmware provides it).

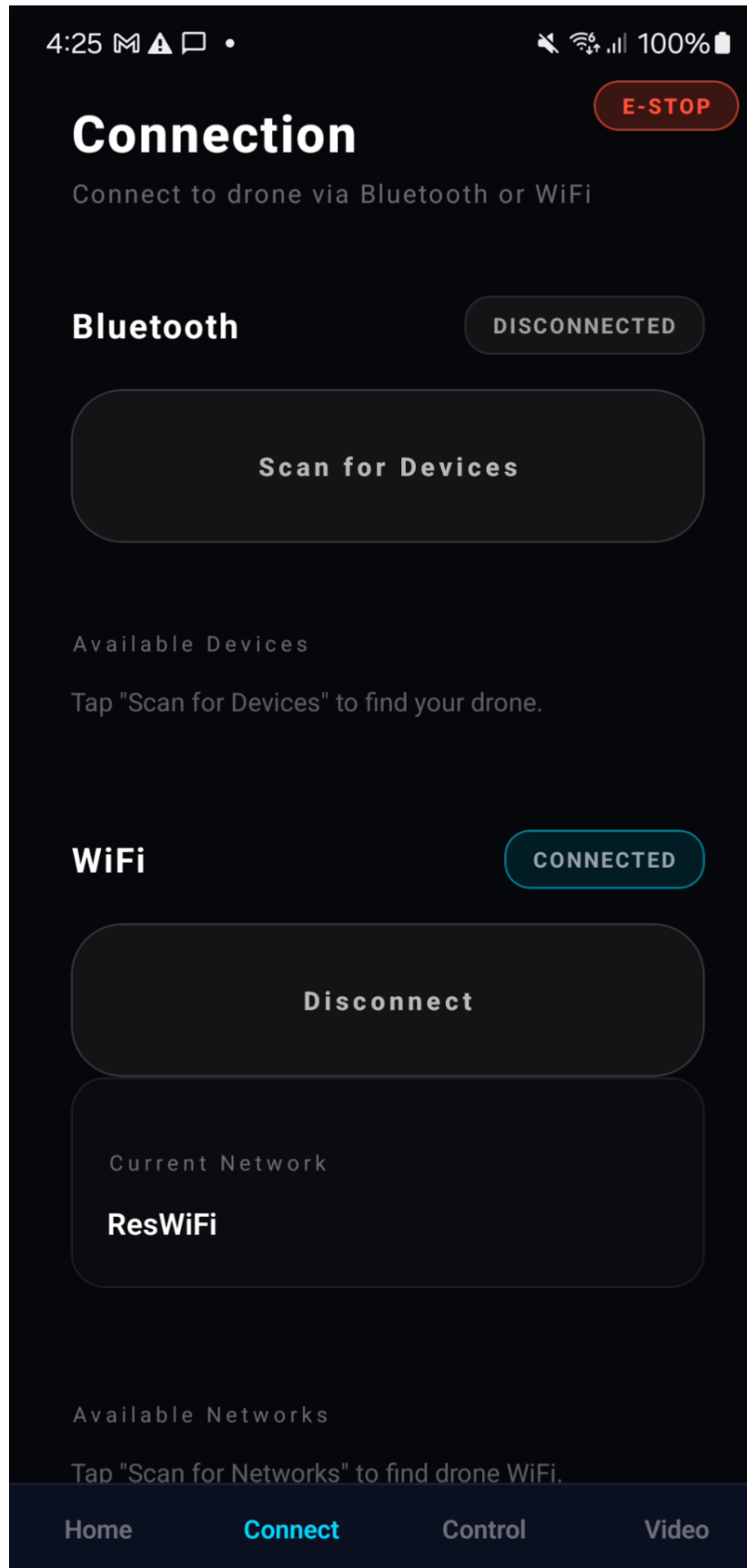


Figure 2: Connect screen

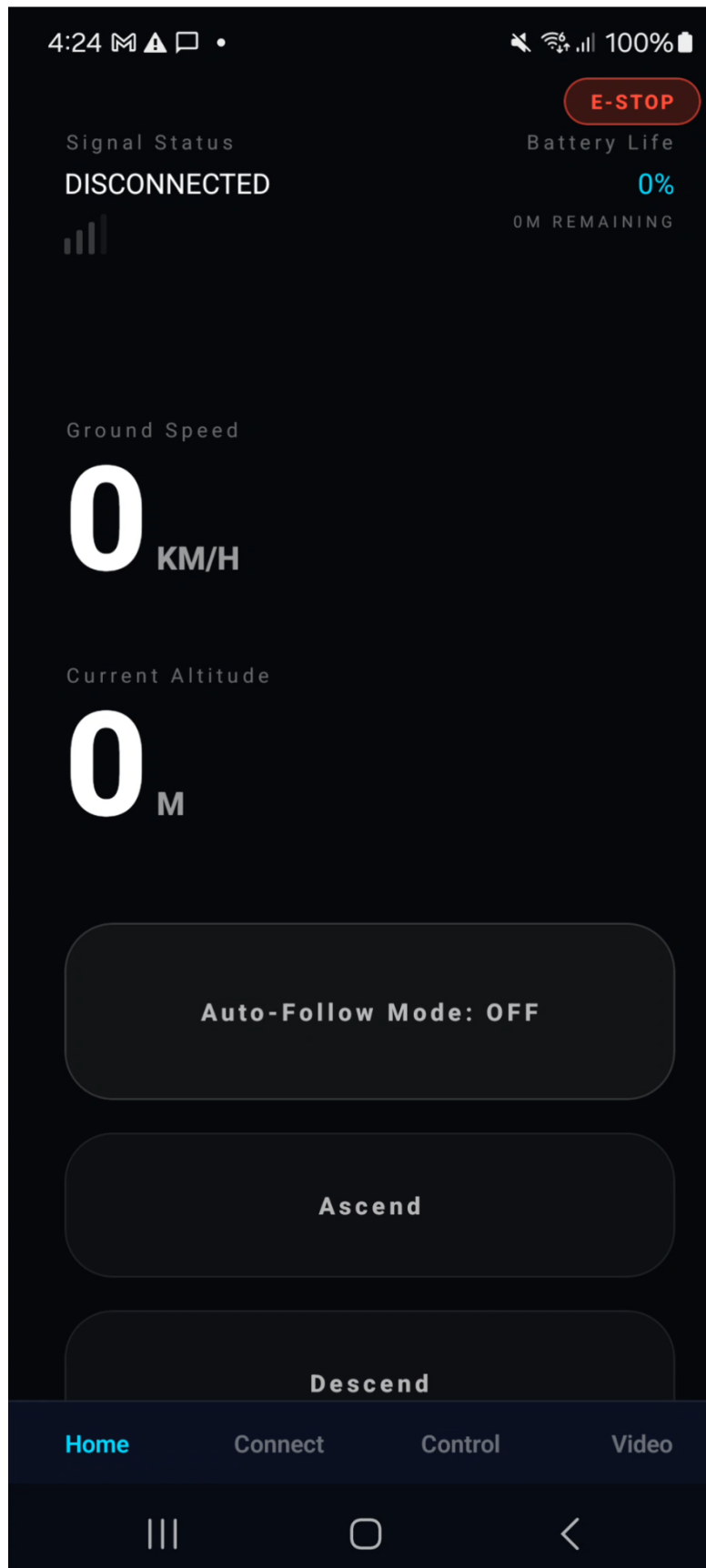


Figure 3: Home screen

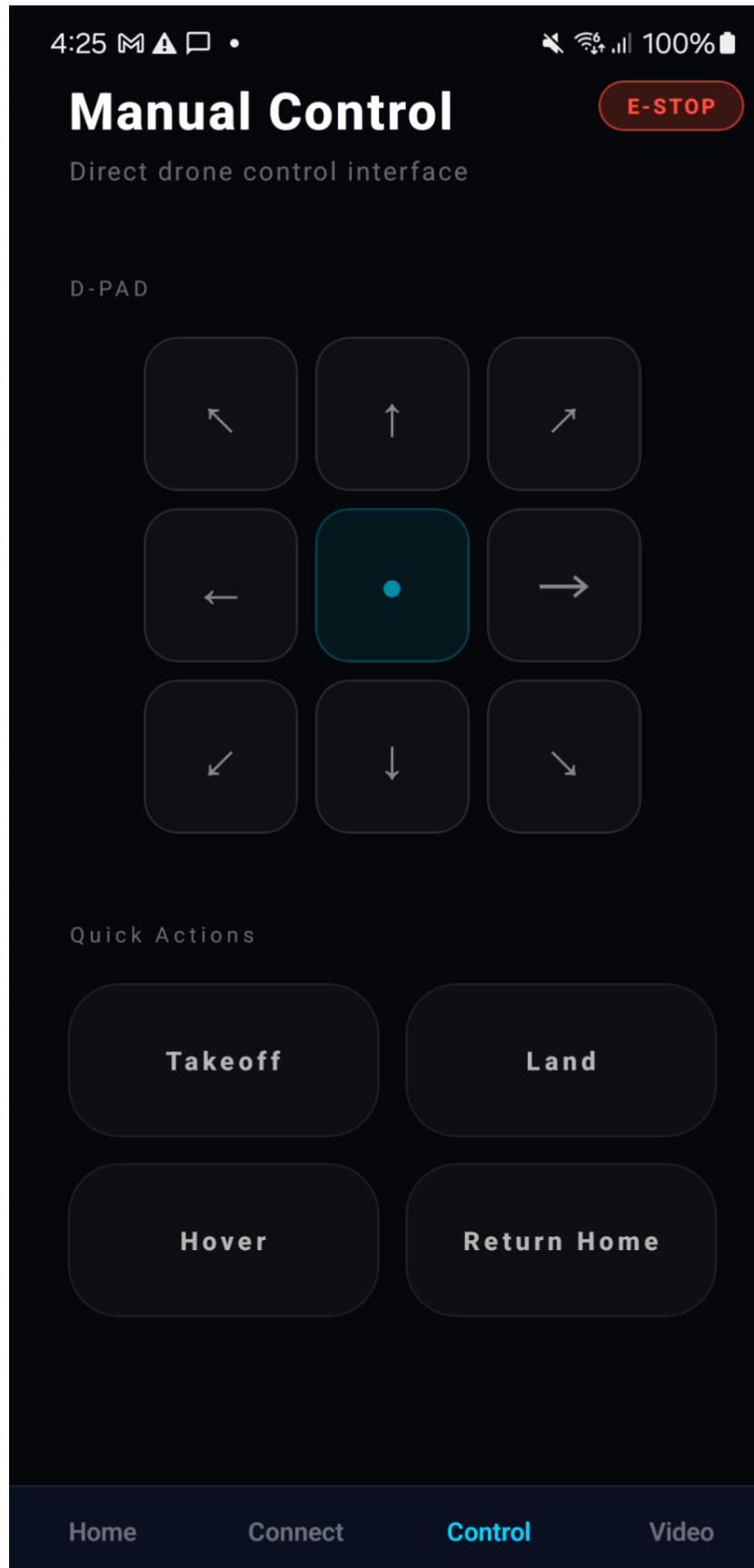


Figure 4: Control screen

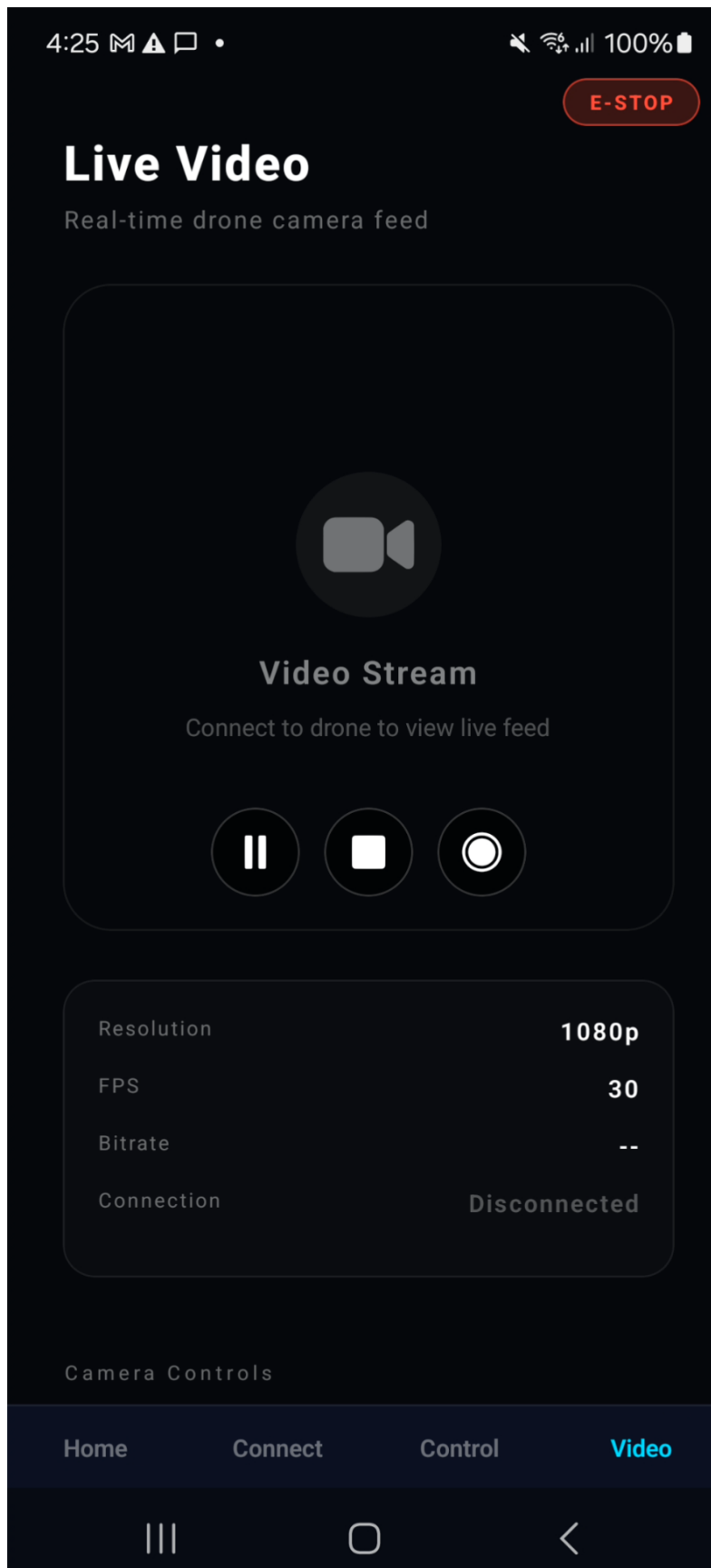


Figure 5: Video screen

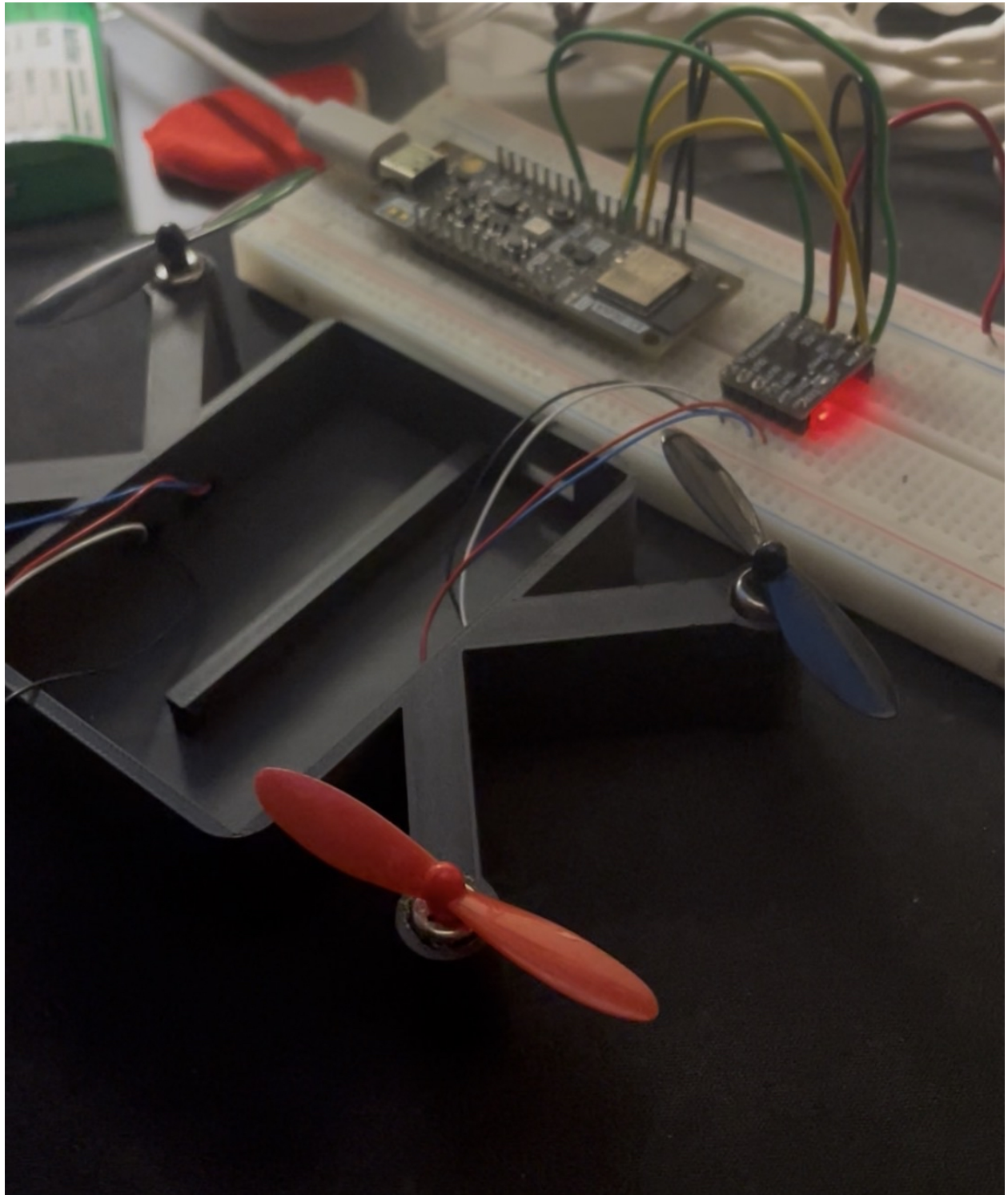
Connect Screen

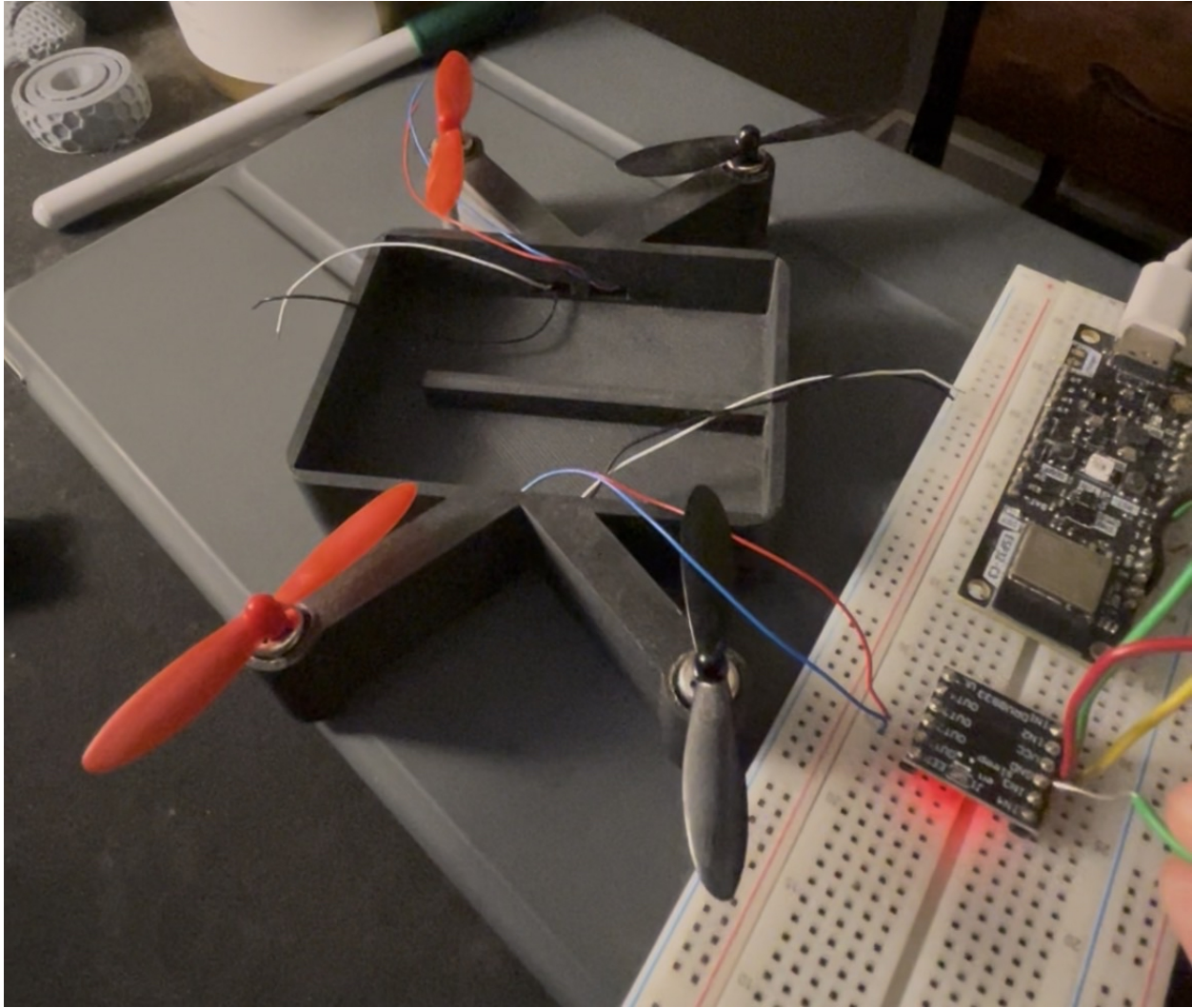
Home Screen

Manual Control

Live Video Feed

Prototype Photos





Testing

The manufacturing test plan in Appendix 3 defines how a fully assembled production drone will be evaluated before it leaves manufacturing. Six tests (MECH-01 through FLT-01) cover visual mechanical inspection, safe power-on and idle current behavior, PCB and sensor self-tests, verification of the control link and command protocol, tethered motor mapping and spin-up checks, and a sample hover/failsafe flight test. Together, these tests are designed to catch structural, electrical, and control issues early and to show that each unit can power on safely, communicate correctly, spin the correct motors in the correct directions, and maintain stable hover under the documented failsafe policy.

Appendix 1 – Problem Formulation

1. Conceptualisations

System concept

The product is conceived as an **autonomous filming drone**: a quadcopter that captures video without requiring the user to pilot it. The user defines what they want to film (e.g. “follow me”,

“orbit this area”, “cover this event”). The drone then flies and films autonomously. Control and monitoring are done via a mobile application over a wireless link. The system is designed so that filming and flying do not depend on drone piloting skills, making it suitable for content creators, sports filming, and event coverage where the focus is on the shot, not on stick control.

Stakeholders and users

- **Content creator / filmmaker.** The primary user is someone who wants to self-document or capture footage without learning to pilot. Examples include solo YouTubers filming themselves (vlogs, activities, tutorials), crews or individuals filming athletes in sports, and event producers using single drones or **larger fleets** to document large events (e.g. sporting events, concerts) from multiple angles.
- **Development and maintenance.** The team (or future maintainers) who develop and update firmware and the mobile app, and who assemble or repair the hardware.
- **Subjects and bystanders.** People being filmed or in the operating environment. Safety and predictable autonomous behaviour matter in shared or crowded spaces.

Main functions (concept level)

1. **Communicate.** Reliable two-way link between user and drone (commands, status, telemetry, and video stream).
2. **Actuate.** Drive the four propellers to achieve lift, orientation, and motion (manual control validated, autonomous flight planned).
3. **Film.** Capture video (and optionally stream it) for self-documentation, sports, or event coverage.
4. **Sense.** Perceive the environment and drone state to support autonomous framing, following, and safety.
5. **Navigate autonomously.** Execute filming behaviours (e.g. follow, orbit, waypoints) and avoid obstacles without manual piloting.
6. **Scale to fleets.** Support multiple drones documenting one event from several angles, coordinated from a single or small number of operators.

Context and scenario

Use spans **solo creators** (e.g. YouTubers filming themselves), **sports** (filming athletes from the air), and **large events** (sporting events, concerts) where one or many drones provide multi-angle coverage without a pilot per drone. The system is intended for environments where autonomous flight is acceptable and where the value is in hands-free filming rather than manual piloting. This conceptualisation describes what the system is and does at a high level, without committing to low-level implementation details.

2. Brainstorming

Brainstorming centred on a few key concepts: the drone should not require piloting skill, it should support self-documentation (e.g. solo creators filming themselves), and it should work without internet. Almost every aspect of the system became a “how do we?” question. The team had limited prior experience in drone flight, mobile app development, and sending data between a phone and embedded hardware, so idea generation focused on identifying options and unknowns rather than assuming solutions. The following themes and ideas came out of those discussions.

1. No piloting skill and self-documentation

- User chooses a high-level goal (e.g. “follow me”, “orbit”, “record this”) rather than flying manually.
- Preset behaviours (follow, orbit, hover) so the user does not need to learn to pilot.
- Self-documentation as the main use case: one person filming themselves without a second operator.
- Operation that works without internet so it is usable in the field or in places with poor connectivity.

2. How do we connect the user to the drone and send data?

- Bluetooth only at first. The team explored BLE for control and pairing with a phone.
- Discovery that BLE alone would not be enough to stream recorded video led to considering alternatives (e.g. WiFi for higher bandwidth when streaming is needed).
- Need to support both control commands and video (or at least video metadata) while keeping “works without internet” in mind.
- Phone app as the primary interface, with only one team member having prior mobile app experience, so app design and communication protocols were open questions.

3. How do we track the subject or know where to fly?

- Computer vision on the drone or on the phone to follow a person or target.
- GPS plus magnetometer for position and orientation (outdoor, where GPS is available).
- The team is still unsure which approach to adopt: computer vision vs GPS and magnetometer, or a combination depending on context.

4. What sensors and parts do we actually need?

- At first the need for many extra parts was not obvious.
- An IMU was assumed necessary and expected to be available on the custom PCB.
- Brainstorming raised the need for a barometer, magnetometer, and possibly a GPS unit.
- Sensors became a major open question: what is strictly necessary for a first version vs what is needed for reliable autonomous behaviour and safety.

3. Decision Tables

Several important design choices were clarified during the early stages of the project. The following decision tables present these choices in a structured way, focusing on communication methods, camera inclusion, and subject tracking approaches.

1. Communication method for prototype and future streaming

Criterion	BLE only (prototype)	WiFi only	Hybrid BLE + WiFi (future)	Cellular or long range
Supports control commands	Yes	Yes	Yes	Yes
Supports video streaming	No (not enough bandwidth)	Yes	Yes	Yes

Criterion	BLE only (prototype)	WiFi only	Hybrid BLE + WiFi (future)	Cellular or long range
Works without internet	Yes	Yes	Yes	Often needs network backend
Power and complexity	Low power and simple	Higher power and more complex	Higher power and more complex	Highest complexity
Fit for first prototype	Chosen for first prototype	Not used yet	Planned for later streaming	Not planned

Decision

We chose BLE first for low-latency control, then added Wi-Fi when bandwidth needs (telemetry at ~5 Hz, GPS JSON, and camera streaming) exceeded BLE capacity. The **finished product uses a hybrid link**: BLE for commands and keep-alive; Wi-Fi (TLS) for telemetry and video. The mobile app implements this split by default (hybrid comms layer).

2. Camera inclusion

Criterion	No camera	Simple fixed camera (chosen)	Stabilised or advanced camera
Supports self documentation	No	Yes	Yes
Hardware and integration complexity	Lowest	Moderate	Highest
Data bandwidth requirements	Very low	Moderate	High
Matches goal of autonomous filming	Does not match goal	Matches goal for first version	Best match but not required yet
Cost	Lowest	Moderate	Highest

Decision

We considered leaving the camera out to reduce complexity, cost, and data rate. After discussion we decided that a camera is essential because the main purpose of the system is autonomous filming and self documentation. We therefore treat a simple fixed camera as required for the first version. More advanced camera systems can be added in later iterations.

3. Subject tracking approach (still under evaluation)

Criterion	Computer vision tracking	GPS plus magnetometer tracking	Manual framing only
Works indoors	Yes if lighting is good	No or limited	Yes
Works outdoors	Yes	Yes	Yes
Extra hardware required	Camera and enough processing	GPS and magnetometer	None beyond basic control

Criterion	Computer vision tracking	GPS plus magnetometer tracking	Manual framing only
Algorithm and implementation effort	High	Moderate	Low
Dependence on internet	Can work offline	Works offline	Works offline
Team experience level	Limited	Limited	Feels most achievable now

Decision

The **first product release** targets **outdoor follow-me using phone GNSS + drone GNSS and magnetometer**: the app computes bearing and distance; the flight controller executes NAV_* motion commands under onboard limits. Computer vision may augment or replace GPS tracking in a later revision. Manual framing and the Control tab remain for setup, recovery, and regulatory testing.

Appendix 2 – Planning

Basic Plan / Gantt Chart Roadmap exported from Jira (Feb–Apr 2026). Epics cover mobile app, hardware procurement, drone firmware, documentation, and Q2 integration work (GPS, navigation, camera stream).

Division of Labor During Prototyping Phase The table below summarizes each team member’s primary areas of contribution from prototyping through the current quarter (Feb–Jun 2026). Work was tracked in Jira; deliverables are reflected in the GitHub monorepo.

Team Member	Primary Contributions
Ethan Liu	Continued mobile app development, including expo-router restructure, BLE fixes, manual controls, and WiFi/hybrid testing. Set up the Jira board, led weekly scrums, and contributed design-document appendix and evaluation updates.
Cameron Dubois	Bootstrapped the mobile app (BLE, Connect UI) and hybrid comms layer. Contributed flight-control bench work, follow-to-phone navigation, and barometer integration across app and firmware.
Darin Rahm	Focused on drone firmware, including ESP32 BLE and WiFi connectivity, GATT command protocol, and soft-AP/TLS integration. Authored the test plan and demonstration tests.
Stephen Wend-Bell	Set up the GitHub repository and documentation pipeline; motor control functions and system diagrams. Integrated GPS/compass bring-up and authored technical design chapters on comms, telemetry, and mobile architecture.
Winnie Wong	Handled parts research and acquisition, PCB schematic, server integration, and campus drone flight guidelines. Authored camera-integration design content and component documentation.
Abhiram Sai Yegalapati	Designed and iterated the drone shell CAD model and wiring layout. Released new frame revisions as board and sensor layout changed.

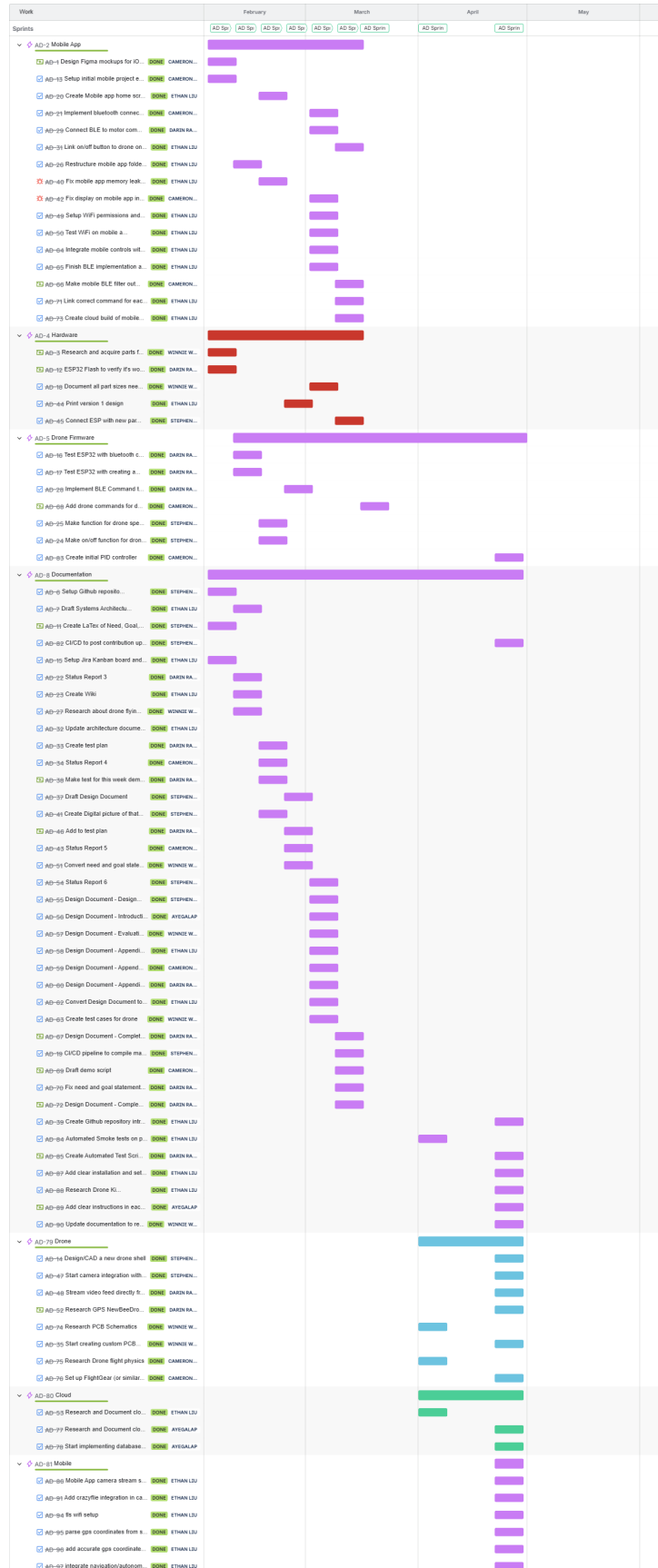


Figure 6: Jira project roadmap

Collaboration We used the following tools and processes to coordinate work:

- **Jira** – We used Jira with Kanban boards and Scrum to manage sprints, bugs, and tasks. Epics and stories were broken down into sprint-sized work, and we tracked progress through columns (e.g., To Do, In Progress, In Review, Done). Sprint planning was held regularly at our first meeting of the week.
- **GitHub** – The repository was used for all code, documentation, and design files.
- **Discord** – Discord served as the main channel for day-to-day messaging, quick questions, meeting coordination, and sharing updates between synchronous meetings.

Appendix 3 – Manufacturing Test Plan & Results

This section defines a generic manufacturing and verification test plan for future engineers who build or maintain production versions of the autonomous drone.

1. Scope & Administrative Details

Scope

- **System under test:** Fully assembled production drones and subassemblies (flight controller PCB, power system, motors).
- **Goal of testing:** Verify that manufactured product meets the project’s functional and safety objectives before being accepted for use (mechanical integrity, power behavior, electronics health, communications, motor mapping, and basic flight/failsafe behavior).
- **Parameters and justification:** Tests focus on parameters that directly impact safety and reliability.
- **Expectations (hypothesis):** Units built to the documented design and assembly process will pass all six manufacturing tests without rework.

Administrative Details

- **Date and location of testing:** To be filled in per production run
 - **Client or organization:** Future owner of the autonomous drone design
 - **Conducting the test:** Manufacturing / Quality Assurance engineers or technicians following this plan.
-

2. Manufacturing Test Definitions

TEST ID: MECH-01 — Visual Mechanical & Assembly Inspection

Scope & goal

- **System under test:** Fully assembled autonomous drone.
- **Goal:** Confirm each assembled drone matches mechanical drawings and is safe to power.

Test design (variables, sampling, apparatus)

- **Type:** Qualitative inspection (yes/no, acceptable/unacceptable).
- **Independent variable:** Individual production unit.
- **Dependent variables:** Pass/fail for each checklist item.
- **Sampling:** Every unit (100% inspection); performed before the unit is powered on for the first time.
- **Apparatus:** Mechanical drawings, good lighting.

Procedure (outline)

1. Place powered-off unit on inspection bench and compare it to the latest mechanical drawing.
2. Verify they are the same.
3. Confirm each propeller is installed in the correct position and orientation; rotate slowly by hand to check for interference.
4. Inspect wiring for pinched insulation, unsecured leads, and exposed conductors; verify connectors are fully seated.
5. Record pass/fail against the checklist; photograph any defects.

Safety & external factors

- No power applied; battery disconnected during inspection.
- Be cautious of sharp edges on printed parts.
- Note any unusual lighting or visibility conditions that could affect inspection quality.

Data collection

- Complete a digital or paper checklist per unit.
- Attach photographs of any defects or borderline conditions.

Pass criteria

- 0 cracked, warped, or obviously damaged structural parts.
- 0 mis-oriented or rubbing propellers.
- 0 exposed conductors or unsafe wiring.

TEST ID: PWR-01 — Power-On & Idle Current Verification

Scope & goal

- **System under test:** Complete drone with production power electronics and firmware.
- **Goal:** Verify power rails, boot behavior, and idle current are within specification.

Test design (variables, sampling, apparatus)

- **Type:** Quantitative electrical test.
- **Independent variable:** Input supply (battery vs. bench supply).
- **Dependent variables:** Boot success, peak inrush current, steady-state idle current, supply voltage.
- **Sampling:** Every unit or defined batch sampling.
- **Apparatus:** Bench supply or production battery, inline current measurement, voltmeter, safe test stand/mat.

Procedure (outline)

1. Connect drone to bench supply (with current limit) or a known-good production battery.
2. Measure and record input voltage and current limit settings.
3. Power on the drone and observe peak inrush current and steady-state idle current once boot is complete.
4. Confirm status indicators or telemetry show that firmware booted successfully (no repeating resets).

Safety & external factors

- Use current limits on bench supply; keep flammable materials away.
- Ensure adequate ventilation; monitor for abnormal heating or smell.
- Record ambient temperature if it may affect current draw.

Data collection

- Log input voltage, current limit, peak inrush current, and steady-state idle current for each unit.
- Record pass/fail plus any anomalies in a shared spreadsheet or database.

Pass criteria

- Drone boots from power-on to ready state.
- Peak inrush current and idle current fall within defined allowable ranges.
- No abnormal heating, smell, or cycling resets during the observation period.

TEST ID: PCB-01 — Electronics Self-Test & Sensor Sanity Check

Scope & goal

- **System under test:** Assembled flight controller PCB with production firmware.
- **Goal:** Confirm that the microcontroller and critical sensors (e.g., IMU, barometer, magnetometer) are present and functional.

Test design (variables, sampling, apparatus)

- **Type:** Mixed qualitative/quantitative functional test.
- **Independent variable:** Unit under test.
- **Dependent variables:** Self-test result code, sensor responsiveness, and basic value ranges.
- **Sampling:** 100 % of PCBs before integration into airframes.
- **Apparatus:** Test jig or debug interface, host PC/machine, self-test script or tool.

Procedure (outline)

1. Connect PCB or fully assembled drone to the test jig or debug port.
2. Trigger the firmware self-test routine and read back the summary status.
3. Query each required sensor at rest and record a short window of data for each.
4. Check that values are within plausible ranges.

Safety & external factors

- Follow Electrostatic Discharge precautions when handling bare PCBs.
- Ensure board is mechanically supported to avoid stressing solder joints.
- Write down the room temperature and local pressure if they change the sensor readings.

Data collection

- Store self-test status codes and raw sensor snapshots with unit serial numbers.
- Keep logs or CSV exports from the test jig software for later analysis.

Pass criteria

- Self-test completes with an “OK” or equivalent status code.
 - All required sensors respond without communication errors.
 - All measured values fall within predefined acceptable ranges for a unit at rest.
-

TEST ID: COMMS-01 — Control Link & Protocol Verification

Scope & goal

- **System under test:** Communication link between ground station (app/controller) and drone, including command/ACK protocol.
- **Goal:** Verify that control commands are correctly received, acknowledged, and constrained by safety rules.

Test design (variables, sampling, apparatus)

- **Type:** Functional communications test.
- **Independent variables:** Command type (e.g., ARM, DISARM, ESTOP, basic movement or throttle commands).
- **Dependent variables:** ACK success/failure codes, internal state transitions, latency.
- **Sampling:** 100 % of units.
- **Apparatus:** Ground station or scripted test client, log capture, safe bench or fixture.

Procedure (outline)

1. Establish a control link between the ground station and the drone on a bench or fixture.
2. Send a scripted sequence of commands including ARM, small thrust changes, ESTOP, DISARM, and at least one “illegal” command (such as thrust while disarmed).
3. Capture command/ACK logs at both ends and record any internal state reported by telemetry.

Safety & external factors

- Perform on a secured bench or fixture; do not allow free-flight for this test.
- Either remove props or limit motor outputs to low test values.
- Minimize radio-frequency interference sources nearby when evaluating latency behavior.

Data collection

- Save timestamped command and ACK logs from both ground station and drone.
- Record any protocol errors, timeouts, and observed latencies per command type.

Pass criteria

- Every test command receives an ACK with matching ID and correct success/error status.
- Illegal commands (e.g., motor command while disarmed) are rejected and do not change motor state.

- ESTOP forces all motor outputs to a safe value within the specified time and requires a deliberate re-arm before further motion.
-

TEST ID: SYS-01 — Motor Mapping & Spin-Up Test (Tethered / No Lift-Off)

Scope & goal

- **System under test:** Fully assembled drone with motors and ESCs, on a tether or fixture.
- **Goal:** Ensure each logical motor drives the correct physical motor, in the correct direction, with smooth low-power response.

Test design (variables, sampling, apparatus)

- **Type:** Functional system-level test.
- **Independent variables:** Commanded motor channel and throttle level.
- **Dependent variables:** Observed motor spin, direction, and current draw.
- **Sampling:** Every unit before first free-flight.
- **Apparatus:** Mechanical fixture or tether, propellers, eye and hearing protection.

Procedure (outline)

1. Secure the drone in a test fixture or tether so it cannot lift off.
2. ensure propellers command each motor individually to low and then medium test speeds.
3. Observe which physical motor spins and confirm rotation direction; listen and feel for abnormal vibration or noise.
4. Record current draw per motor channel if available.

Safety & external factors

- Always secure the drone in a fixture and keep hands clear of rotating parts.
- Use eye and hearing protection when motors are spinning (if needed).
- Note any unusual ambient vibration or mechanical noise in the test area.

Data collection

- Record pass/fail for motor ID and direction per channel.
- Log current draw and any abnormal observations for units that need rework.

Pass criteria

- Logical motor IDs match the correct physical motors for all channels.
 - All motors spin in the intended direction with no abnormal vibration or noise.
 - Measured current per motor lies within expected limits for the test speeds.
-

TEST ID: FLT-01 — Production Acceptance Hover & Failsafe Test (Sample Units)

Scope & goal

- **System under test:** Representative production drones in a safe test environment.
- **Goal:** Confirm that units can maintain a stable hover and execute basic failsafe behavior (e.g., link loss response) as specified.

Test design (variables, sampling, apparatus)

- **Type:** System-level flight test.
- **Independent variables:** Unit sample, commanded altitude, control link state.
- **Dependent variables:** Position/altitude deviation during hover, behavior when link is lost or degraded.
- **Sampling:** Any unit that passes the previous tests moves onto this one.
- **Apparatus:** Safe test area or netted flight cage, ground station, means to log telemetry and link state.

Procedure (outline)

1. In a designated test area, take off and climb to a fixed test altitude (for example, 2–3 m above ground).
2. Command a hover and maintain it for a defined time window while recording position and altitude deviations.
3. Intentionally drop or degrade the control link according to the documented failsafe scenario and observe behavior until landing or timeout.

Safety & external factors

- Conduct tests only in a safe, controlled area (e.g., netted flight cage or open test range) with observers briefed on emergency procedures.
- Comply with all relevant safety and airspace regulations.
- Record environmental conditions such as wind speed, temperature, and lighting.

Data collection

- Log telemetry (position, altitude, link status) for the full duration of the test.
- Capture notes or video for any unexpected behaviors during hover or failsafe response.

Pass criteria

- During hover, position and altitude remain within defined tolerances/ranges for the duration of the test.
- When the link is lost or degraded, the drone follows the documented failsafe policy (e.g., hover then land, or return-to-home then land) and does not exhibit uncontrolled motion or fly-away behavior.

Appendix 4 – Review

One paragraph from each team member

Ethan Liu

Our integration went well largely because we kept everything in a monorepo — mobile app, firmware, and documentation all in one place. That made it easier to coordinate changes, run tests across subsystems, and keep the design document aligned with the actual codebase. If I were to do it again, I would devote more time to the drone hardware side. Our effort was fairly evenly split between hardware and firmware, but in retrospect I would skew toward hardware earlier: soldering, mechanical assembly, and hands-on debugging of the physical drone to unblock firmware and flight testing sooner.

Stephen Wend-Bell

In this quarter we made a lot of good progress on our prototype and design. We went from nothing to having a working mobile app, working bluetooth connection to the drone, and all four motors spinning in response to the mobile app. One that hindered my progress at least was my understanding of what was required of us in the class, I originally thought we were just building our design. It wasn't until like week 5 that I realized we're just building a prototype, and the main focus of the class is just the design. After we understood this, our work became more focused and we made more progress.

Cameron Dubois

This quarter I shifted from the mobile app toward the drone's flight and sensing systems, which is the direction I said I wanted to take after last quarter. I brought up the flight control firmware on our board — IMU reading, the PID control loop, and motor mixing — and built a hardware-in-the-loop simulation so we could tune and test control logic without always needing a tethered airframe. On the app side I added the follow-to-phone navigation module and a follow-mock that translates navigation intent into the drone's command set, which gave us a working app-to-firmware path to demonstrate. I also integrated the barometer for altitude telemetry. The main thing still ahead is closing the loop so the flight controller fully executes those navigation commands in real flight rather than logging them, and if I were pacing myself again I would have started the firmware work earlier instead of carrying so much of the app focus into the start of the quarter.

Winnie Wong

I think one thing that went well during our project was the progress our team made on different parts of the drone. We were able to work on multiple components at the same time, such as the mobile app and the drone prototype, which helped us keep moving forward. Everyone in the group focused on different aspects of the project, and that division of work helped us make steady progress overall. If I were to do this again, I would try to better understand the main requirements earlier in the quarter, especially the design expectations and documentation that were required later on. Knowing that earlier would have helped me focus more on those parts of the project from the beginning and manage my time more effectively.

Darin Rahm

Looking back on what I worked on, which was the firmware and test development, one thing that went well was getting a reliable BLE link. From there my teammate was able to take my code and their existing motor control code, and combine them together so the team could see and hear the prototype respond in real time. I also established a reliable wifi connection but we did not need to use it yet. Where I would change my approach is in how I framed my initial testing because my test plans were written for our initial prototype, rather than for the manufactured product we decided to create. If I were to do this again, I would focus more from the perspective of a production line engineer. I would start by designing manufacturing-level tests with clear pass/fail criteria, along with repeatable procedures, and consistent data logging. Having this shift in mindset would have made it easier to tie our day-to-day debugging directly to the long-term reliability requirements of the finished drone. I also realized that our need statement, design objective, and goal statement were never really solidified in my mind, so when I worked on the design document and prepared for the test demo I wasn't consistently checking that my tests and decisions traced back to those three statements and all connected.

Abhiram Sai Yegalapati

When I started working on this project, I had a lot of ideas on what could be integrated in this

design. With a design such as a drone, there could be a lot of possibilities on how to add features. After figuring out what type of design we fixed on, then implementing some of the features will take more time than I realized. Before thinking more about that, I was assigned on making a drone shell that was necessary for the demo at the end of the quarter. So, coming with multiple ideas on what type of drone shell would work would constantly change. I made changes everytime there was a change in dimension as new parts kept getting added. In fact, we still don't have a finalized drone for next quarter yet because we have more features to implement. Maybe, I could have used some help on what features we would have finally implemented, but that's nearly impossible as features kept changing. My next attention is going to work on integrating more features such as a charger and camera into the drone shell and also make the drone fly.